Post-Quantum TLS Readiness Assessment Tool

Project No : 10

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1.Problem Statement

Most websites today use classical cryptographic algorithms such as RSA and ECC to secure TLS communications. With the rapid advancement of quantum computing, these algorithms may become vulnerable in the future. While existing tools can analyze current TLS security, they do not assess how prepared a website is for the transition to post-quantum cryptography. This project aims to develop a tool that evaluates the TLS configuration of websites and provides a Post-Quantum Readiness Score along with recommendations to improve future security.

2.Objectives

The main objectives of this project are:

To identify the cryptographic algorithms and key sizes used by websites.

To evaluate the post-quantum readiness of a website using a scoring model.

To generate security recommendations based on the analysis.

To present the results through a simple web-based interface.

3.Background and Motivation

Transport Layer Security (TLS) is the standard protocol used to secure communication between users and websites by encrypting data transmitted over the internet. Most websites currently rely on classical cryptographic algorithms such as RSA and Elliptic Curve Cryptography (ECC) to establish secure connections.

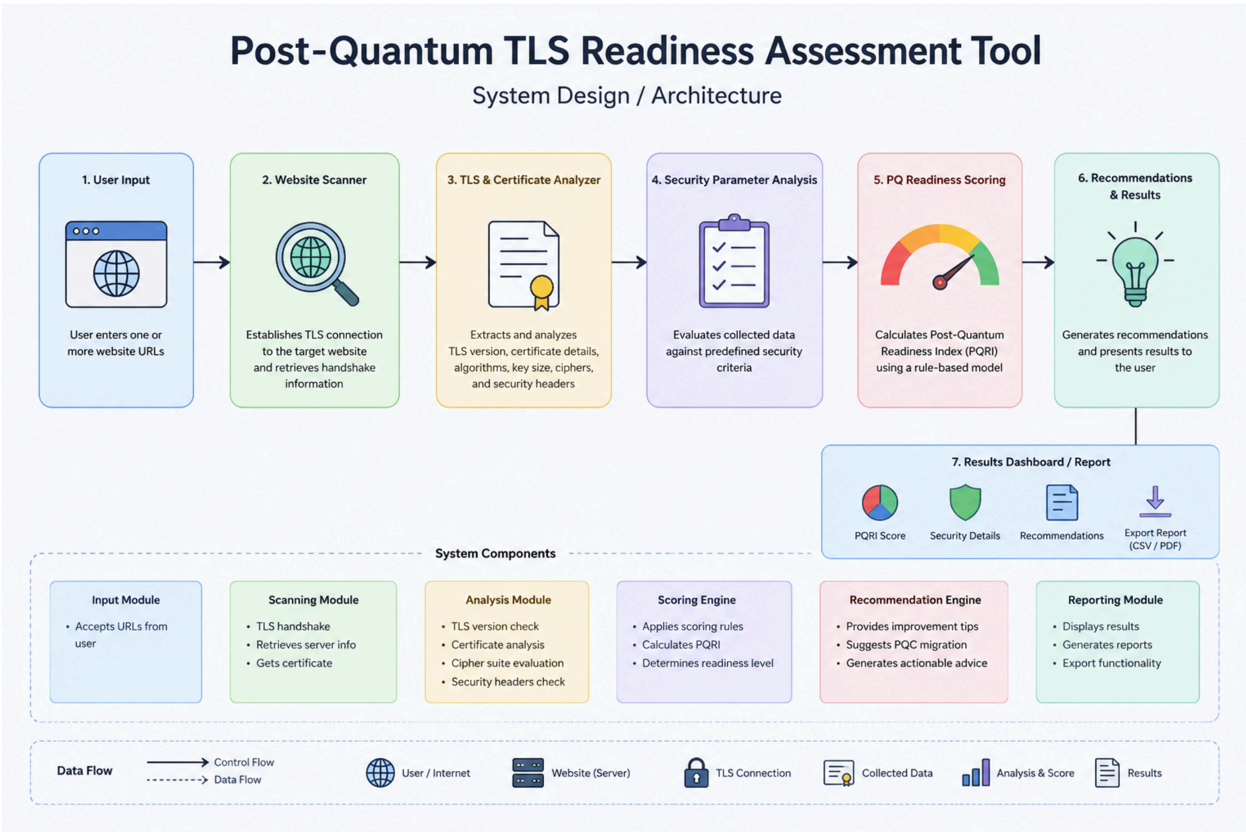
However, the rapid advancement of quantum computing poses a potential threat to these widely used cryptographic algorithms. As organizations begin preparing for the transition to Post-Quantum Cryptography (PQC), there is an increasing need to assess whether existing web infrastructure is ready for this shift.

The motivation behind this project is to develop a practical tool that analyzes the TLS configuration of websites, evaluates their readiness for post-quantum migration, and provides recommendations to help improve their future security posture.

4.Proposed Solution

The proposed system is a Python-based application that analyzes the TLS security configuration of websites and evaluates their readiness for the transition to post-quantum cryptography.
The application will establish a secure connection with the target website and collect publicly available information such as the TLS version, certificate details, public-key algorithm, key size, certificate validity period, and selected HTTP security headers. Based on these parameters, the system will calculate a Post-Quantum Readiness Score using a predefined scoring model.
The final analysis will include recommendations to improve the website's readiness for future post-quantum cryptographic standards. The results will be displayed through a simple web interface and can also be exported as a report.

5.System Design / Architecture



6.Algorithms / Techniques

The project will make use of standard networking and security techniques to analyze websites and evaluate their readiness for post-quantum migration.

Cryptographic Algorithms Analyzed

·RSA

·Elliptic Curve Cryptography (ECC)

Post-Quantum Reference Algorithms

·ML-KEM (Kyber)

·ML-DSA (Dilithium)

Techniques Used

·TLS Handshake Analysis

·SSL/TLS Certificate Analysis

·X.509 Certificate Parsing

·Rule-Based Risk Scoring

·Recommendation Generation

Tools and Frameworks

·Python

·SSL and Socket Programming

·Cryptography Library

·Requests

·Pandas

·Matplotlib

·Streamlit

7.Implementation Plan

The project will be developed using Python as the primary programming language. Initially, the application will establish secure TLS connections with websites and collect publicly available certificate and security information. The collected data will then be analyzed to identify the cryptographic algorithms used, certificate validity, TLS version, and other relevant security parameters.

Based on this analysis, a Post-Quantum Readiness Score will be calculated using a rule-based scoring model. The application will also generate recommendations to improve the website's readiness for future post-quantum cryptographic migration. Finally, the results will be presented through a simple web interface that allows users to easily understand the overall security assessment.

Software Requirements

·Python 3.x

·Visual Studio Code

·OpenSSL

·Git

·Streamlit

Hardware Requirements

·Laptop/Desktop

·Internet Connection

Development Methodology

·Modular and Incremental Development

8. Expected Outcomes

The expected outcome of this project is a functional tool that can analyze the TLS security configuration of websites and assess their readiness for post-quantum cryptography. The system will generate a Post-Quantum Readiness Score, provide recommendations for improving security, and present the results through a simple web interface. The project will also help users understand the potential impact of quantum computing on existing cryptographic systems and encourage early adoption of quantum-safe security practices.